

Names: _____ Period: _____ Date: _____

LAB #15: MODELING MOLECULES & POLYATOMIC IONS

This lab will help you understand the information needed to make a Lewis-dot structure, how to construct the activity and how to confirm whether or not the structure is reasonable.

Before you attempt to sketch the Lewis-dot structure for a molecule or a polyatomic ion, you should:

1. Determine the number of valence electrons in each of the elements in the molecule or ion, using the periodic table.
2. Inspect the molecule to determine whether or not it has a charge.
3. Determine the total number of valence electrons in the molecule or ion.

When trying to make the sketch, you should:

1. Make single atom or carbon atom the central atom.
2. Usually, the remaining atoms are placed around the central atom. The atoms should be placed around the central atom so that the unpaired valence electrons pair up to form covalent bond(s) with the central atom.

When you think you have a reasonable structure:

1. Check to make sure every non-metal atom has an octet. Exception: hydrogen (2 valence electrons in total)
2. If it doesn't have an octet, you may:
 - a) Need to make double or triple bonds
 - b) Have forgotten to add or remove valence electrons if it has a charge
 - c) Have simply not placed the correct number of valence electrons in the structure, or
 - d) The central atom is one of the exceptions to the octet rule (Elements: As, P, S, Cl, Br, and I can form compounds with 10 or 12 valence electrons)

DIRECTION: Use the following color key in the Lewis Structure for each of the molecules or ions.

Atom	Color	Atom	Color
F, Br	Red	N, P	Brown
C	Green	S	Purple
O	Yellow	As	Orange
H	Blue	bonds, lone pairs	Black

Chemical Formula (Write the student's name who built this)	Number of Valence Electrons (show work)	Lewis Structure (use color key in instructions)	Lone Pair (Put "N/A" if there is no central atom)	BED	RED	Molecule Shape	Hybridization	STAMP
F₂CO								
C₃H₈								
H₂CO								
HCN								
CO₂								

Chemical Formula (Write the student's name who built this)	Number of Valence Electrons (show work)	Lewis Structure (use color key in instructions)	Lone Pair (Put "N/A" if there is no central atom)	BED	RED	Molecule Shape	Hybridization	STAMP
SO₂								
C₃H₆								
PH₄⁺¹								
NCS⁻¹								
N₂								

Chemical Formula (Write the student's name who built this)	Number of Valence Electrons (show work)	Lewis Structure (use color key in instructions)	Lone Pair (Put "N/A" if there is no central atom)	BED	RED	Molecule Shape	Hybridization	STAMP
SO₄⁻²								
CO₃⁻²								
C₅H₁₂O								
PBr₃								
NO₃⁻								